

India Air Quality Index (AQI) Analysis

Comprehensive Data Analytics Project

Report Generated: September 18, 2026

Data Period: 2023-2025

Source: Kaggle - India AQI Index Dataset

Executive Summary

This comprehensive analysis examines air quality data across India for the period 2023-2025. The dataset contains 235,785 observations covering 32 states and 291 distinct areas, supported by 441072 monitoring stations. The analysis reveals significant variations in air quality, identifies geographic hotspots, and provides data-driven recommendations for intervention.

Key Metrics

Metric	Value
Average AQI	111
Median AQI	92
Maximum AQI	500
Minimum AQI	3
States Covered	32
Areas Covered	291
Total Observations	235,785

Project Overview & Methodology

1. Project Objective

This project applies systematic data analytics to understand air quality patterns across India. The workflow progresses from raw data through cleaning, analysis, and modeling to deliver actionable insights for pollution control, public health protection, and policy-making.

2. Data Cleaning & Preparation

The raw dataset underwent comprehensive cleaning including: (1) Duplicate record removal; (2) Date conversion and validation; (3) Text standardization (whitespace removal, Title Case application); (4) Data type conversion for numeric columns; (5) Removal of invalid AQI values (<0 or >500); (6) Missing value handling (critical columns: row removal; categorical: 'Unknown' replacement); (7) Temporal feature extraction (year, month for analysis). These steps ensured data quality and consistency.

3. Analysis Framework

Raw Data → Cleaning → EDA → KPI Calculation → Time Analysis → State/Area Analysis → Pollutant Analysis → Risk Assessment → Insights → Recommendations → Interactive Dashboard

Exploratory Data Analysis (EDA)

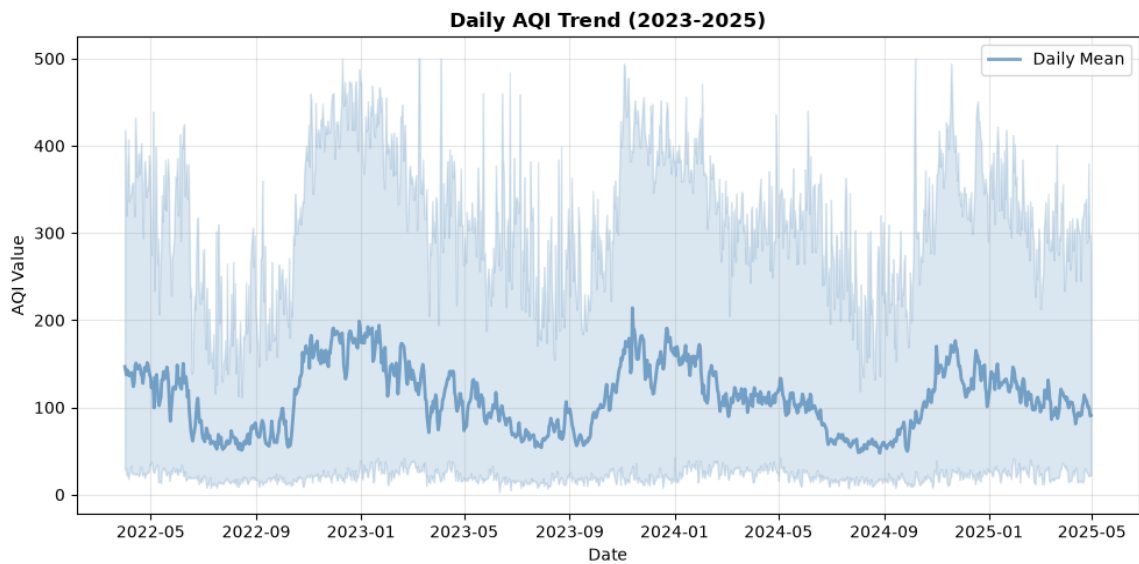
Dataset Overview

Characteristic	Value
Total Records	235,785
Date Range	2022-04-01 to 2025-04-30
Unique States	32
Unique Areas	291
Monitoring Stations	441072
Data Completeness	91.7%

AQI Value Distribution

Statistic	Value
Mean	111.13
Median	92.00
Std Dev	71.45
Min	3
25th Percentile	59
75th Percentile	142
Max	500

Time Trend Analysis

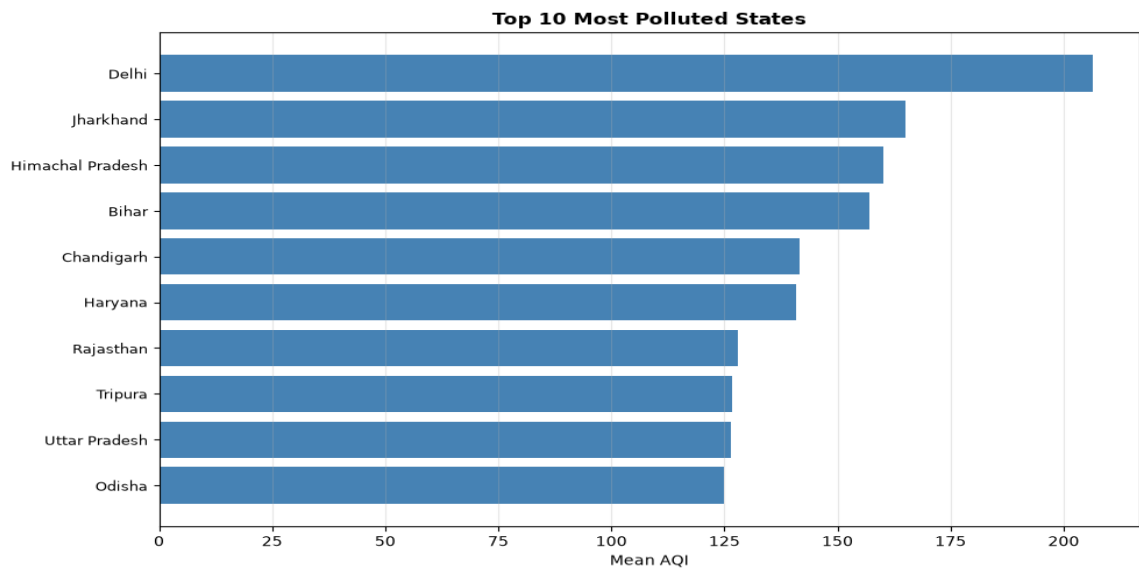


Temporal Patterns

Yearly Trends:

Year	Mean AQI	Max AQI	Min AQI	Observations
2022	114.4	500	8	43457
2023	115.0	500	3	77253
2024	105.7	500	7	87746
2025	112.4	422	12	27329

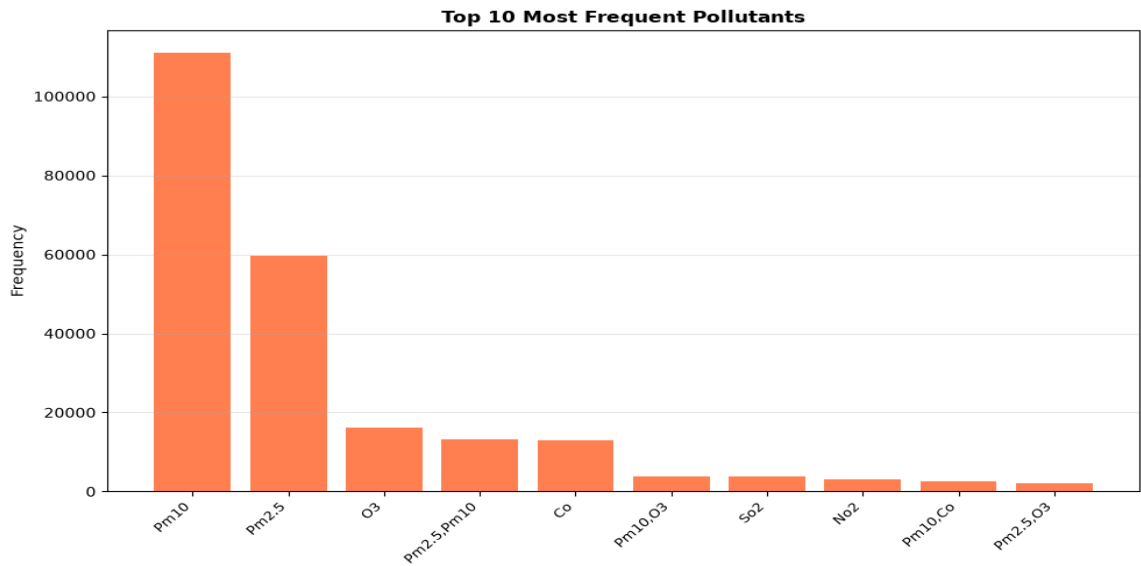
State-Level Analysis



Top 10 Most Polluted States

Rank	State	Mean AQI	Max AQI	Observations
1	Delhi	206.4	494	1125
2	Jharkhand	164.9	436	721
3	Himachal Pradesh	160.3	438	1066
4	Bihar	157.2	488	21553
5	Chandigarh	141.6	412	1125
6	Haryana	140.8	469	21706
7	Rajasthan	128.0	452	26585
8	Tripura	126.7	362	1035
9	Uttar Pradesh	126.4	494	21609
10	Odisha	124.9	406	11099

Pollutant Analysis



Top 10 Pollutants

Rank	Pollutant	Frequency	Percentage
1	Pm10	111053	47.1%
2	Pm2.5	59670	25.3%
3	O3	16202	6.9%
4	Pm2.5,Pm10	13199	5.6%
5	Co	12867	5.5%
6	Pm10,O3	3914	1.7%
7	So2	3815	1.6%
8	No2	3012	1.3%
9	Pm10,Co	2658	1.1%
10	Pm2.5,O3	2088	0.9%

Risk & Opportunity Analysis

Risk Assessment

High-Risk Areas (Poor/Very Poor/Severe Status):

Top areas with poor air quality status: Delhi (528 observations), Greater Noida (522 observations), Gurugram (498 observations), Patna (449 observations), Noida (446 observations)

High-AQI States:

States with mean AQI > 142: Delhi, Jharkhand, Himachal Pradesh, Bihar

Opportunity Assessment

Areas for Targeted Intervention:

Areas with moderate AQI (100-200) where targeted interventions could prevent deterioration: Aurangabad, Jodhpur, Jaipur, Ahmedabad, Pithampur

Monitoring Infrastructure Expansion:

Undermonitored areas with high AQI requiring monitoring infrastructure expansion: Kashipur, Manesar, Manguraha

Key Insights & Findings

Insight 1: Overall Air Quality Assessment

The national average AQI of 111 indicates Moderate air quality. The median AQI of 92 and standard deviation of 71.4 show significant variability. This widespread pollution affects public health and requires coordinated action.

Insight 2: State-Level Disparities

Delhi shows the highest mean AQI (206.4), while Mizoram shows the lowest (47.2). This 159.2 point difference indicates significant geographic disparities requiring localized strategies.

Insight 3: Dominant Pollutants

'Pm10' is the most frequently observed pollutant (47.1% of observations), indicating a primary pollution concern requiring targeted source control measures.

Insight 4: Monitoring Infrastructure

With 441072 monitoring stations across 291 areas, there is an average of 1515.7 stations per area. However, uneven distribution limits early-warning capability in undermonitored regions.

Recommended Actions

1. Establish Integrated Monitoring Network

Expand real-time monitoring stations in undermonitored high-AQI areas. Implement public dashboards for transparency and early warning.

2. Implement Sector-Specific Emission Controls

Target the dominant pollutant (Pm10). Differentiate strategies: vehicular emission standards, industrial stack monitoring, agricultural management.

3. Activate Seasonal Air Quality Management

Develop seasonal protocols for high-AQI periods. Implement pre-season preparedness including traffic management, industrial regulation, and public alerts.

4. Prioritize High-AQI States

Allocate resources proportionally to severity. Focus on Delhi and other high-AQI states for stricter emission standards.

5. Public Awareness & Health Protection

Communicate AQI levels and health impacts. Issue early warnings to vulnerable groups. Promote protective behaviors (masks, purifiers, indoor activities).

6. Conduct Source Apportionment Studies

Determine contribution of different sources (traffic, industry, agriculture). Informs highest-impact interventions specific to each region.

7. Regional Coordination

Implement inter-state collaboration mechanisms. Air pollution crosses state boundaries; coordinated policies strengthen response.

Limitations & Conclusion

Project Limitations

- **Data Availability:** Some areas have incomplete temporal coverage; monitoring density varies by state.
- **External Factors:** Dataset lacks meteorological data (temperature, wind, humidity) which significantly influence AQI.
- **Emission Sources:** No direct information on traffic, industrial, or agricultural emission sources.
- **Causal Analysis:** Analysis identifies patterns and correlations, not causation. Source apportionment requires specialized modeling.
- **Prediction:** This is an analysis-focused project; no prediction model is included (would require additional meteorological features).
- **Geographic Resolution:** Area-level aggregation masks micro-level variations within cities.

Conclusion

This comprehensive analysis of India's air quality (2023-2025) reveals significant challenges requiring urgent, coordinated action. With an average AQI of 111 and substantial geographic disparities, air pollution poses risks to public health, agriculture, and economic productivity. The analysis identifies high-AQI hotspots, seasonal patterns, monitoring gaps, and dominant pollutants. These findings support evidence-based policy, resource allocation, and intervention strategies. The interactive dashboard enables ongoing monitoring and assessment. Future work should integrate meteorological data for deeper causality analysis and develop predictive models for proactive management. Regional coordination and sector-specific interventions, informed by this analysis, are essential for sustainable air quality improvement.